
Pin Map

Pin Name	Pin	Pin Name	Pin
S1	1	B1	2
A1	3	S0	4
A0	5	B0	6
CIN	7	COUT	13
S3	14	B3	15
A3	16	S2	17
A2	18	B2	19

Global Parameters

Parameter	Value
VCC Pin	20
GND Pin	8
VCC Voltage	2.5, 3.3, 4.0, 4.5, 5
Input Low	0.75, 0.99, 1.2, 1.35, 1.5
Input High	1.75, 2.31, 2.8, 3.15, 3.5
Output Low	0.3, 0.3, 0.3, 0.3, 0.3
Output High	2.2, 3.0, 3.6, 4.1, 4.7

Tests: **IC PASS 12/12**

*A=0, B=0, CIN=0 -> S=0: **PASS***

Inputs			VCC	Outputs/Results									
A3, A2, A1, A0	B3, B2, B1, B0	CIN		COUT		S3		S2		S1		S0	
0b0000	0b0000	0b0	2.5	0	0.0 (0)	0	0.0 (0)	0	0.0 (0)	0	0.0 (0)	0	0.0 (0)
			3.3		0.0 (0)		0.0 (0)		0.0 (0)		0.0 (0)		0.0 (0)
			4.0		0.0 (0)		0.0 (0)		0.0 (0)		0.0 (0)		0.0 (0)
			4.5		0.0 (0)		0.0 (0)		0.0 (0)		0.0 (0)		0.0 (0)
			5		0.0 (0)		0.0 (0)		0.0 (0)		0.0 (0)		0.0 (0)

A=0, B=0, CIN=1 -> S=1: PASS

Inputs			VCC	Outputs/Results									
A3, A2, A1, A0	B3, B2, B1, B0	CIN		COUT		S3		S2		S1		S0	
0b0000	0b0000	0b1	2.5	0	0.0 (0)	0	0.0 (0)	0	0.0 (0)	0	0.0 (0)	1	2.37 (1)
			3.3		0.0 (0)		0.0 (0)		0.0 (0)		0.0 (0)		3.15 (1)
			4.0		0.0 (0)		0.0 (0)		0.0 (0)		0.0 (0)		3.81 (1)
			4.5		0.0 (0)		0.0 (0)		0.0 (0)		0.0 (0)		4.43 (1)
			5		0.0 (0)		0.0 (0)		0.0 (0)		0.0 (0)		4.85 (1)

A=1, B=0, CIN=0 -> S=1: PASS

Inputs			VCC	Outputs/Results									
A3, A2, A1, A0	B3, B2, B1, B0	CIN		COUT		S3		S2		S1		S0	
0b0001	0b0000	0b0	2.5	0	0.0 (0)	0	0.0 (0)	0	0.0 (0)	0	0.0 (0)	1	2.37 (1)
			3.3		0.0 (0)		0.0 (0)		0.0 (0)		0.0 (0)		3.15 (1)
			4.0		0.0 (0)		0.0 (0)		0.0 (0)		0.0 (0)		3.81 (1)
			4.5		0.0 (0)		0.0 (0)		0.0 (0)		0.0 (0)		4.43 (1)
			5		0.0 (0)		0.0 (0)		0.0 (0)		0.0 (0)		4.86 (1)

A=0, B=1, CIN=0 -> S=1: PASS

Inputs			VCC	Outputs/Results									
A3, A2, A1, A0	B3, B2, B1, B0	CIN		COUT		S3		S2		S1		S0	
0b0000	0b0001	0b0	2.5	0	0.0 (0)	0	0.0 (0)	0	0.0 (0)	0	0.0 (0)	1	2.37 (1)
			3.3		0.0 (0)		0.0 (0)		0.0 (0)		0.0 (0)		3.15 (1)
			4.0		0.0 (0)		0.0 (0)		0.0 (0)		0.0 (0)		3.81 (1)
			4.5		0.0 (0)		0.0 (0)		0.0 (0)		0.0 (0)		4.43 (1)
			5		0.0 (0)		0.0 (0)		0.0 (0)		0.0 (0)		4.86 (1)

A=3, B=5, CIN=0 -> S=8: PASS

Inputs			VCC	Outputs/Results											
A3, A2, A1, A0	B3, B2, B1, B0	CIN		COUT		S3		S2		S1		S0			
0b0011	0b0101	0b0	2.5	0	1	0	0.0 (0)	0	2.37 (1)	0	0.0 (0)	0	0.0 (0)	0	0.0 (0)
			3.3				0.0 (0)		3.15 (1)		0.0 (0)		0.0 (0)		0.0 (0)
			4.0				0.0 (0)		3.81 (1)		0.0 (0)		0.0 (0)		0.0 (0)
			4.5				0.0 (0)		4.43 (1)		0.0 (0)		0.0 (0)		0.0 (0)
			5				0.0 (0)		4.85 (1)		0.0 (0)		0.0 (0)		0.0 (0)

A=15, B=1, CIN=0 -> S=16: PASS

Inputs			VCC	Outputs/Results									
A3, A2, A1, A0	B3, B2, B1, B0	CIN		COUT		S3		S2		S1		S0	
0b1111	0b0001	0b0	2.5	1	2.37 (1)	0	0.0 (0)	0	0.0 (0)	0	0.0 (0)	0	0.0 (0)
			3.3		3.15 (1)		0.0 (0)		0.0 (0)		0.0 (0)		0.0 (0)
			4.0		3.81 (1)		0.0 (0)		0.0 (0)		0.0 (0)		0.0 (0)
			4.5		4.43 (1)		0.0 (0)		0.0 (0)		0.0 (0)		0.0 (0)
			5		4.85 (1)		0.0 (0)		0.0 (0)		0.0 (0)		0.0 (0)

A=15, B=15, CIN=0 -> S=30: PASS

Inputs			VCC	Outputs/Results									
A3, A2, A1, A0	B3, B2, B1, B0	CIN		COUT		S3		S2		S1		S0	
0b1111	0b1111	0b0	2.5	1	2.37 (1)	1	2.37 (1)	1	2.37 (1)	1	2.37 (1)	0	0.0 (0)
			3.3		3.15 (1)		3.15 (1)		3.15 (1)		3.15 (1)		0.0 (0)
			4.0		3.81 (1)		3.81 (1)		3.81 (1)		3.81 (1)		0.0 (0)
			4.5		4.43 (1)		4.43 (1)		4.43 (1)		4.43 (1)		0.0 (0)
			5		4.85 (1)		4.85 (1)		4.85 (1)		4.85 (1)		0.0 (0)

A=15, B=15, CIN=1 -> S=31: PASS

Inputs			VCC	Outputs/Results									
A3, A2, A1, A0	B3, B2, B1, B0	CIN		COUT		S3		S2		S1		S0	
0b1111	0b1111	0b1	2.5	1	2.37 (1)	1	2.37 (1)	1	2.37 (1)	1	2.37 (1)	1	2.37 (1)
			3.3		3.15 (1)		3.15 (1)		3.15 (1)		3.15 (1)		
			4.0		3.81 (1)		3.81 (1)		3.81 (1)		3.81 (1)		
			4.5		4.43 (1)		4.43 (1)		4.43 (1)		4.43 (1)		
			5		4.85 (1)		4.85 (1)		4.85 (1)		4.85 (1)		

A=10, B=5, CIN=0 -> S=15: PASS

Inputs			VCC	Outputs/Results									
A3, A2, A1, A0	B3, B2, B1, B0	CIN		COUT		S3		S2		S1		S0	
0b1010	0b0101	0b0	2.5	0	0.0 (0)	1	2.37 (1)	1	2.37 (1)	1	2.37 (1)	1	2.37 (1)
			3.3		0.0 (0)		3.15 (1)		3.15 (1)		3.15 (1)		
			4.0		0.0 (0)		3.81 (1)		3.81 (1)		3.81 (1)		
			4.5		0.0 (0)		4.44 (1)		4.44 (1)		4.44 (1)		
			5		0.0 (0)		4.85 (1)		4.86 (1)		4.85 (1)		4.85 (1)

A=10, B=10, CIN=1 -> S=21: PASS

Inputs			VCC	Outputs/Results									
A3, A2, A1, A0	B3, B2, B1, B0	CIN		COUT		S3		S2		S1		S0	
0b1010	0b1010	0b1	2.5	1	2.37 (1)	0	0.0 (0)	1	2.37 (1)	0	0.0 (0)	1	2.37 (1)
			3.3		3.15 (1)		0.0 (0)		3.15 (1)		0.0 (0)		3.15 (1)
			4.0		3.81 (1)		0.0 (0)		3.81 (1)		0.0 (0)		3.81 (1)
			4.5		4.43 (1)		0.0 (0)		4.43 (1)		0.0 (0)		4.43 (1)
			5		4.85 (1)		0.0 (0)		4.86 (1)		0.0 (0)		4.86 (1)

A=7, B=1, CIN=0 -> S=21: PASS

Inputs			VCC	Outputs/Results									
A3, A2, A1, A0	B3, B2, B1, B0	CIN		COUT		S3		S2		S1		S0	
0b0111	0b0001	0b0	2.5	0	1	2.37 (1)	0	0.0 (0)	0	0.0 (0)	0	0.0 (0)	
			3.3					0.0 (0)		0.0 (0)		0.0 (0)	
			4.0					0.0 (0)		0.0 (0)		0.0 (0)	
			4.5					0.0 (0)		0.0 (0)		0.0 (0)	
			5					0.0 (0)		0.0 (0)		0.0 (0)	

A=7, B=3, CIN=1 -> S=21: PASS

Inputs			VCC	Outputs/Results									
A3, A2, A1, A0	B3, B2, B1, B0	CIN		COUT		S3		S2		S1		S0	
0b1100	0b0011	0b1	2.5	1	2.37 (1)	0	0.0 (0)	0	0.0 (0)	0	0.0 (0)	0	0.0 (0)
			3.3		3.15 (1)		0.0 (0)		0.0 (0)		0.0 (0)		
			4.0		3.81 (1)		0.0 (0)		0.0 (0)		0.0 (0)		
			4.5		4.43 (1)		0.0 (0)		0.0 (0)		0.0 (0)		
			5		4.85 (1)		0.0 (0)		0.0 (0)		0.0 (0)		